

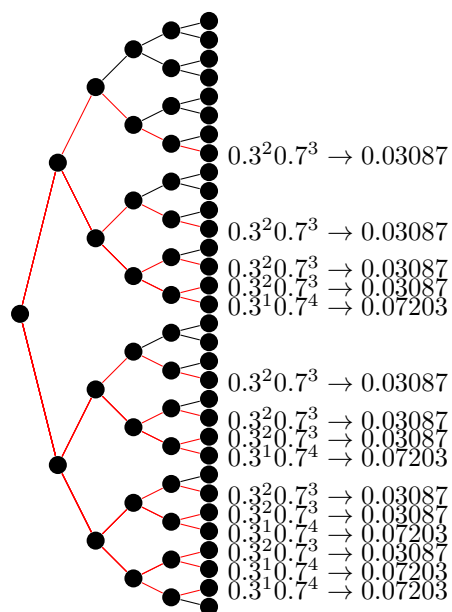
Homework 6

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Exercise 1

The following tree represents the possible outcomes of flipping a coin with a bias of $\theta = 0.3$ five times:



The probability that the frequency deviates from the true bias by at most $\epsilon = 0.1$ is $0.03087 \cdot 10 + 0.07203 \cdot 5 = 0.66885$.

Using the formula:

$$\sum_{x=1}^2 \binom{5}{x} 0.3^x (1-0.3)^{5-x} = \binom{5}{1} 0.3^1 0.7^4 + \binom{5}{2} 0.3^2 0.7^3$$

$$= 0.66885$$

Exercise 2

The probability that the frequency of landing heads in 40 tosses is 0.05-close to the true bias of 1/4 is:

$$\begin{aligned} P_{1/4,40}(|\text{frequency of landing heads} - 1/4| \leq 0.05) &= \sum_{x: \left| \frac{x}{40} - \frac{1}{4} \right| \leq 0.05} \text{Binom}_{1/4,40}(x) \\ &= \sum_{x=8}^{12} \text{Binom}_{1/4,40}(x) \\ &= \sum_{x=8}^{12} \frac{40!}{x!(40-x)!} \left(\frac{1}{4}\right)^x \left(\frac{3}{4}\right)^{40-x} \\ &\approx 0.6389 \end{aligned}$$

Exercise 3

$$\begin{aligned}0.95 &\leq 1 - \frac{1}{4n(0.01)^2} \\ \frac{1}{4n(0.01)^2} &\leq 0.05 \\ 50000 &\leq n\end{aligned}$$

When estimating the bias of the coin by the observed frequency under the IID assumption, the estimation procedure has at least a 95% probability of having an error less than 0.01 when the coin is tossed at least 50000 times.